

Sai Manikanta Karnati

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years building scalable microservices and distributed systems across Python, Java, TypeScript, and Go. Skilled in AWS, Azure, and GCP cloud-native delivery, B2B platform engineering, and event-driven backend design. Reduced production issue resolution time by 40% and boosted transaction performance 35% while supporting recruiting, onboarding, and compensation product teams.

SKILLS

Python - Java - Go - TypeScript - JavaScript - Microservices - Distributed Systems - Spring Boot - Node.js - Angular - REST - GraphQL - AWS - Azure - GCP - Kubernetes - Docker - Terraform - Apache Kafka - CI/CD - SQL - DynamoDB - MongoDB - Redis - B2B

EXPERIENCE

Software Engineer - Vanguard

08/2025 - present

- Designed and developed scalable backend microservices in Python and Java Spring Boot, enabling modular service-oriented architecture and independent deployments across distributed systems.
- Engineered Go and Python data-processing pipelines and automation scripts, cutting manual operational effort ~50% and improving backend processing efficiency.
- Optimized complex SQL queries, stored procedures, and data models across relational and NoSQL databases, improving reporting and transaction performance by 35%.
- Built high-throughput event-driven systems using DynamoDB Streams, Python/Node.js Lambda functions, and SNS/SQS, ensuring 99.99% reliability for B2B service workflows.
- Developed responsive Angular SPAs in TypeScript with RxJS and Reactive Forms, integrated via RESTful and GraphQL APIs for seamless frontend-backend delivery.
- Deployed cloud-native applications on AWS (Lambda, ECS, EC2, S3, RDS, DynamoDB) and Azure, adding Splunk and Actuator observability that reduced production issue resolution time by 40%.

Associate Software Engineer - Xebia

01/2022 - 12/2023

- Developed backend applications in Python, Java Spring Boot, Go, and Node.js, delivering scalable RESTful APIs and microservices for B2B clients.
- Built Python automation scripts for data transformation, validation, and backend task orchestration across distributed systems.
- Designed and optimized SQL queries, joins, views, and stored procedures, improving API response times by 35%.
- Implemented event-driven architecture with Apache Kafka, reducing inter-service dependency failures by 35%.
- Created and documented RESTful and GraphQL APIs using Swagger/OpenAPI, enforcing versioning and backward compatibility for partner teams.
- Automated infrastructure provisioning with Terraform on AWS and GCP, and containerized services using Docker and Kubernetes to improve CI/CD deployment efficiency by ~40%.

Engineer Trainee - Xebia

01/2020 - 12/2021

- Developed enterprise applications using Angular and Python/Java backend services for transactional business platforms.
- Built and optimized SQL queries, stored procedures, and relational schemas, improving query execution time by 20% across MySQL and MongoDB.
- Developed RESTful APIs and Angular TypeScript modules with Redux and Reactive Forms, documenting service contracts via Swagger/OpenAPI.
- Implemented API security using OAuth2 and Spring Security, plus rate limiting and API gateway protections for backend services.
- Increased unit test coverage to 95% using JUnit and Python-based test automation tools.

EDUCATION

Master of Science in Computer Science - University of North Carolina at Charlotte

CERTIFICATIONS

- AWS Certified Developer Associate
- MongoDB Certified Developer
- Infosys Certified Software Programmer
- NPTEL Certified Java Developer
- AWS Academy Graduate - AWS Academy Data Engineering